



UNIVERSITY OF THE PUNJAB

Fifth Semester – 2019

Examination: B.S. 4 Years Program

Roll No.

PAPER: Econometrics-I

Course Code: ECON-302 Part – II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

- Q2. Explain the following briefly. 4 x 5 = 20
- Coefficient of Determination
 - Perfect and partial multicollinearity
 - Indirect Least Squares
 - Order and Rank conditions
 - Generalized Least Square
- Q3. Consider the following regression output: 10
- $$\hat{y}_i = 0.2033 + 0.6560X_i$$
- $$se = (0.0976) (0.1961)$$
- $$r^2 = 0.397 \quad RSS = 0.0544 \quad ESS = 0.0358$$
- where Y = labor force participation rate (LFPR) of women in 1972 and X = LFPR of women in 1968. The regression results were obtained from a sample of 19 cities in the United States.
- How do you interpret this regression?
 - Test the hypothesis: $H_0: \beta_1 = 1$ against $H_1: \beta_1 > 1$. Which test do you use? And why? What are the underlying assumptions of the test(s) you use?
 - Suppose that the LFPR in 1968 was 0.58 (or 58 percent). On the basis of the regression results given above, what is the mean LFPR in 1972? Establish a 95% confidence interval for the mean prediction.
 - How would you test the hypothesis that the error term in the population regression is normally distributed? Show the necessary calculations.
- Q4. Consider the sets of data given in the following two tables: 10
- | Y | X2 | X3 |
|---|----|----|
| 1 | 2 | 4 |
| 2 | 0 | 2 |
| 3 | 4 | 12 |
| 4 | 6 | 0 |
| 5 | 8 | 16 |
- | Y | X2 | X3 |
|---|----|----|
| 1 | 2 | 4 |
| 2 | 0 | 2 |
| 3 | 4 | 0 |
| 4 | 6 | 12 |
| 5 | 8 | 16 |
- The only difference between the two tables is that the third and fourth values of X_3 are interchanged.
- Regress Y on X_2 and X_3 in both tables, obtaining the usual OLS output.
 - What difference do you observe in the two regressions? And what accounts for this difference?
- Q5. From the data for 46 states in the United States for 1992, Baltagi obtained the following regression results: 10
- $$\log \hat{C} = 4.30 - 1.34 \log P + 0.17 \log Y$$
- $$se = (0.91) (0.32) (0.20) \quad R^2 = 0.27$$
- where C = cigarette consumption, packs per year
 P = real price per pack
 Y = real disposable income per capita
- What is the elasticity of demand for cigarettes with respect to price? Is it statistically significant? If so, is it statistically different from one?
 - What is the income elasticity of demand for cigarettes? Is it statistically significant? If not, what might be the reasons for it?
 - How would you retrieve R^2 from the adjusted R^2 given above?

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q2. Give short answers of the following:

(4x5=20)

- i. Use of t-test and F-test
- ii. Assumptions of Ordinary Least Square Method
- iii. Causes of Multi-collinearity
- iv. Time Series and Cross sectional data
- v. Omitted Variable Bias

Give brief answers of the followings.

(3x10=30)

Q3.	From the following data estimate partial regression coefficients, coefficient of determination. Also provide respective interpretation. <table> <tr> <td>Y</td><td>100</td><td>75</td><td>80</td><td>70</td><td>50</td><td>70</td><td>85</td><td>100</td><td>110</td><td>60</td></tr> <tr> <td>X₂</td><td>5</td><td>7</td><td>6</td><td>7</td><td>8</td><td>7</td><td>4</td><td>4</td><td>3</td><td>9</td></tr> <tr> <td>X₃</td><td>1000</td><td>600</td><td>1100</td><td>500</td><td>300</td><td>400</td><td>1300</td><td>1100</td><td>1400</td><td>300</td></tr> </table>	Y	100	75	80	70	50	70	85	100	110	60	X ₂	5	7	6	7	8	7	4	4	3	9	X ₃	1000	600	1100	500	300	400	1300	1100	1400	300
Y	100	75	80	70	50	70	85	100	110	60																								
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X ₃	1000	600	1100	500	300	400	1300	1100	1400	300																								
Q4.	Define hetroskedasticity. How to find the presence of hetroskedasticity? What are the consequences of hetroskedasticity and its remedies?																																	
Q5.	The estimated model pertaining to child mortality (CM) on per capita GNP (PGNP) and the female literacy rate (FLR) for a sample of 64 countries is reported below along respective standard errors in parenthesis. $\hat{CM}_i = 263.6416 - 0.0056PGNP_i - 2.2316FLR_i$ $Se = (11.5932), (0.0019), (0.2099)$ - Is coefficient of PGNP statistically significant? - Is coefficient of FLR statistically significant? - Are both coefficients statistically significant?																																	



Q.1. Answer the following short questions:

(6x5=30)

1. What is Chow test? Explain its usage.
2. How correlation analysis differs from regression analysis?
3. What are the assumptions of classical linear regression model?
4. What is the OLS criterion of best fitted line?
5. Explain R^2 and adjusted R^2 .
6. Differentiate between high and perfect multicollinearity.

Answer the following questions.

(3x10=30)

Question 2

An economist is trying to establish the extent of the relationship between the amount earned by a taxpayer during the year and his charitable contributions which are tax deductible. A random selection of 9 tax returns are analyzed and the results are tabulated below. The income and contributions are expressed in lakhs of rupees.

Y (Contributions)	2.0	2.5	2.7	2.5	3.8	4.2	4.3	4.4	4.45
X (Income)	25	29	32	32	40	43	50	55	57

- I. Compute values of regression coefficients and interpret estimated parameters.
- II. For a person earning Rs. 65 lakhs per year, how much contribution do you expect him to make?

Question 3

Define Multicollinearity and discuss its consequences, detection methods and remedial measures.

Question 4.

From the following data find regression coefficients and test individual statistical significance of the coefficients.

Year	2000	2001	2002	2003	2004	2005	2006	2007
Qd (Y)	100	75	80	70	50	65	90	100
Price (X1)	15	17	16	16	18	17	15	14
Income (X2)	1000	600	1200	500	300	400	1300	1100

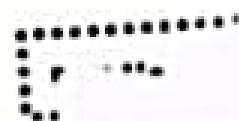


UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Fall 2022

Paper: Econometrics – I

Course Code: ECON-305-A



Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

1. What is the OLS criterion of best fitted line?
2. Explain R^2 and adjusted R^2 .
3. Differentiate between high and perfect multicollinearity.
4. Discuss time series, cross sectional and panel data with examples.
5. Differentiate between sample regression function and population regression function.
6. Discuss statistical properties of OLS estimators.

Answer the following questions.

(3x10=30)

Q.2.

An economist is trying to establish the extent of the relationship between the amount earned by a taxpayer during the year and his charitable contributions which are tax deductible. A random selection of 9 tax returns are analyzed and the results are tabulated below. The income and contributions are expressed in lakhs of rupees.

Y (Contributions)	2.0	2.5	2.7	2.5	3.8	4.2	4.3	4.4	4.85
X (Income)	25	29	32	33	40	43	50	55	57

- I. Compute values of regression coefficients and interpret estimated parameters.
- II. For a person earning Rs. 65 lakhs per year, how much contribution do you expect him to make?

Q.3.

Define auto-correlation and discuss its consequences, detection methods and remedial measures.

Q.4.

From the following data find regression coefficients and test individual statistical significance of the coefficients.

Year	2000	2001	2002	2003	2004	2005	2006	2007
Gd (Y)	100	75	80	70	50	65	90	100
Price (X1)	15	17	15	16	18	17	15	14
Income (X2)	1000	600	1200	500	300	400	1300	1100



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

1. Discuss time series, cross sectional and panel data with examples.
2. What do you mean by coefficient of determination? What are its properties?
3. Discuss statistical properties of OLS estimators.
4. Discuss consequences of auto-correlation for OLS estimators.
5. Differentiate between OLS and GLS (Generalized Least Square) estimators.
6. Why do we need regression analysis? Why not simply use the mean value of the regressand as its best value?

Answer the following questions.

(3x10=30)

Question 2

a) Consider the following OLS regression model:

$$\ln y_t = \alpha + \beta \ln x_t + \varepsilon_t$$

Where $y_t \rightarrow$ Real food expenditure, $x_t \rightarrow$ Real food advertising expenditure, $\ln \rightarrow$ log values are used to compress the scale. The model was estimated using quarterly data 1983-1992 and following results were obtained.

$$\ln y_t = 8.236 + 0.0175 \ln x_t$$

$$R^2 = 0.35$$

- I. Interpret estimated regression coefficients.
- II. Comment on nature of regression.
- III. Comment on value of R^2 . State reasons pertaining to low value of R-square.

b) Income and Consumption expenditure (x and y respectively) are given for 20 families. Given the data,

- I. Estimate consumption function.
- II. Indicate value of MPC (marginal propensity to consume).
- III. Indicate consumption level at zero income and give economic interpretations of your regression results.

$$\sum x = 191200 \quad \sum y = 175300 \quad \sum x^2 = 2,165,180,000 \quad \sum y^2 = 1,813,469,000$$

$$\sum xy = 1,971,670,000$$

Question 3. Define heteroskedasticity and discuss its consequences, detection methods and remedial measures.

Question 4.

From the following data find regression coefficients and test individual statistical significance of the coefficients.

Year	2000	2001	2002	2003	2004	2005	2006	2007
Oil [Y]	100	75	80	70	50	65	90	100
Price [X1]	15	17	16	16	18	17	15	14
Income [X2]	1000	600	1200	500	300	400	1100	1100



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- 4 x 5
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Table 1		
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5	8	16

Table 2		
Y	X2	X3
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The only difference between the two tables is that the third and fourth values of X_3 are interchanged.

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